

Inpatient Monitoring and Assessment Tool

CMT: CMT-0014 (SR)

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Introduction

During hospital inpatient care, the case manager monitors the patient's progress against the Optimal Recovery Course (ORC) for the specific diagnosis or condition found in Inpatient & Surgical Care or General Recovery Care. Additional care needs, listed in the hospital care management section of Inpatient & Surgical Care, direct the case manager to ensure that diagnostic tests, treatments, procedures, and consultations are completed in a timely manner. In addition, the case manager monitors the patient's responses to therapy and care. The case manager consolidates all the assessment information and coordinates transition of care to the next level of care.(1)(2) The discharge planning section of Inpatient & Surgical Care contains information to assist the case manager in ensuring key considerations are followed to complete a safe and timely discharge.

This tool focuses on assessment and monitoring of a hospital patient. It provides the inpatient case manager general information to provide ongoing care planning for the medical, surgical, pediatric, and behavioral health patient. It is not intended to duplicate or replace the information found in the Optimal Recovery Guideline (ORG) for a specific diagnosis.

Medical

- **General status assessment and monitoring**
 - Admission assessments completed, including(3):
 - Conditions present on admission
 - Recent medical history
 - Treatments received prior to hospitalization
 - Diagnostic testing completed recently
 - Fall risk, if applicable(4)
 - Nutrition risk
 - Infection control screen and possible procedures required(5)
 - Discharge options
 - Readmission risk(6)
 - Extended stay risk
 - ADL and IADL ability assessed. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#).
 - Psychosocial functioning assessed. See Psychosocial Assessment [SR](#).
 - Social determinants of health assessed. See Social Determinants of Health Screening Tool [SR](#).
 - Medication reconciliation completed. See Medication Reconciliation Tool [SR](#).
 - Pain status assessed. See Pain Management [SR](#).
- **Neurologic status assessment and monitoring(7)**
 - Complete assessments, including:
 - Delirium risk(8)
 - Mental status examination
 - Cognitive impairment
 - Speech and swallowing ability
 - Monitor for:
 - Orientation, confusion, and thought process changes
 - Headaches

- Motor function, gait, and strength changes
- Radiculopathy from nerve root or spinal cord compression
- Weakness or numbness: facial or extremity, unilateral or bilateral
- Seizure activity
- Hearing or visual changes
- Speech difficulties
- Syncope
- **Cardiovascular status assessment and monitoring(9)**
 - Complete assessments, including:
 - Hemodynamic stability
 - Blood pressure
 - Heart rate and rhythm, including arrhythmias or palpitations
 - Orthostasis
 - Telemetry
 - Presence of implantable cardiac defibrillator or pacemaker
 - Monitor for:
 - Cardiac enzyme changes
 - Chest pain or angina
 - Cyanosis
 - Fatigue
 - Perfusion deficiency, particularly to lower extremities (eg, pedal pulses, color, temperature, or sensation of extremities)
 - Weight gain and dependent edema
 - Infusion therapy needs
 - Transfusions with packed red blood cells or platelets
 - Presence of vascular access device
- **Respiratory status assessment and monitoring(10)(11)**
 - Complete assessments, including:
 - Lung sounds
 - Crackles
 - Rhonchi
 - Wheezes
 - Respiratory rate, rhythm, quality, and effort, including:
 - Cough and characteristics of sputum
 - Oxygen saturation by pulse oximetry
 - Peak expiratory flow rate
 - Intubation and ventilation parameters
 - Use of supplemental oxygen
 - Monitor for:
 - Respiratory status changes (eg, drop in pulse oximetry, sputum, peak flow rate)
 - Cyanosis
 - Orthopnea
 - Dyspnea
 - Intubation and ventilation parameter changes
 - Need for suctioning, chest physiotherapy, or nebulizers to improve pulmonary status
- **Gastrointestinal status assessment and monitoring(12)(13)**
 - Complete assessments, including:
 - Fluid or electrolyte imbalance
 - Weight and body mass index
 - Bowel function, sounds, and movements: quantity, color, frequency, and consistency
 - Nausea and vomiting
 - Abdominal pain, tenderness, distention, or rigidity
 - Nutritional status(14):
 - Risk for malnutrition[A]
 - Ability to tolerate oral fluids and foods
 - Fluid or electrolyte imbalance
 - Enteral or parenteral nutrition
 - Weight and body mass index
 - Risk for dehydration
 - Monitor for:
 - Abdominal pain, distention, ascites, tenderness, or masses
 - Nausea or vomiting
 - Jaundice

- Bowel function changes (eg, constipation, diarrhea, melena)
- **Renal status assessment and monitoring(16)**
 - Complete assessments, including:
 - Renal function (eg, intake and output, creatinine clearance)
 - Medications used that affect renal system:
 - Polypharmacy
 - Nephrotoxic medicines (eg, gentamicin, captopril, methotrexate, NSAIDs, and contrast media)
 - Urinary appearance and output
 - Pain or burning with urination
 - Urgency, frequency, or hesitancy
 - Hematuria or urine discoloration
 - Stones retrieved on straining urine
 - Urethral catheter, if present
 - Dialysis information:
 - If temporary venous access, observe site for patency, redness, swelling, or drainage.
 - If peritoneal dialysis, observe site for redness, swelling, or drainage.
 - For patients attending outpatient hemodialysis, notify center of patient's hospitalization.
 - Monitor for:
 - Renal function changes (eg, change in intake or output, creatinine clearance)
 - Flank pain (with or without radiation to groin)
 - Urinary tract infection
 - Chills
 - Fever
 - Dysuria
 - Frequency
 - Urinary tract obstruction
 - Frequent urination
 - Oliguria or anuria
 - Weak stream
 - Nephrostomy drainage for ureteral obstruction
 - Catheter care and management (ie, Foley, suprapubic, or straight)
- **Integumentary status assessment and monitoring(17)**
 - Complete assessments, including:
 - Skin status, including cyanosis, jaundice, diaphoresis, or poor skin turgor
 - Skin integrity, including inflammation, rash, open wounds, pressure injuries, or bruising. See Wounds Due to Surgery, Injury, or Pressure  SR.
 - Lesions
 - Monitor for:
 - Skin status or integrity changes
 - Signs of skin or wound infection
 - Redness, warmth, swelling, tenderness, or drainage
 - Cellulitis - acute, nonnecrotizing inflammation of dermis and hypodermis(18)
 - Abscesses and exudates
 - Necrotizing fasciitis involves deep soft tissues, causing increasingly severe localized pain.
- **Vision and auditory status assessment and monitoring(19)**
 - Complete assessments, including:
 - Visual acuity
 - Use of glasses, contact lenses, or other visual aids (eg, magnifying glass or lens)
 - Ocular and nonocular history[B]
 - Hearing ability
 - Use of hearing aids
 - Tinnitus, nystagmus, or vertigo
 - Monitor for:
 - Visual or auditory changes (eg, decrease in ability)
- **Discharge planning.** See Discharge Planning Tool  SR.(20)
 - Discharge planning initiated
 - Discharge planning completed

Surgical

- **General status assessment and monitoring**
 - Complete admission assessments, including(3)(21):

- Conditions present on admission
 - Diagnostic testing, if needed
 - Fall risk, if applicable
 - Nutrition risk
 - Infection control screen and possible procedures required(5)
 - Discharge options
 - Readmission risk(6)
 - Extended stay risk
 - Ambulation impairment screening(8)
 - Surgical risk
- ADL and IADL ability assessed. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#).
- Psychosocial functioning assessed. See Psychosocial Assessment [SR](#).(22)
- Social determinants of health assessed. See Social Determinants of Health Screening Tool [SR](#).
- Medication reconciliation completed. See Medication Reconciliation Tool [SR](#).
- Pain status assessed. See Pain Management [SR](#).
- **Neurologic status assessment and monitoring(8)(21)**
 - Complete assessments, including:
 - Level of consciousness
 - Mentation
 - Cognitive function
 - Memory
 - Motor function, strength, gait, balance, and range of motion
 - Speech and swallowing ability
 - Monitor for:
 - Orientation, confusion, and thought process changes
 - Motor function, gait, and strength changes
 - Diminished sensation
 - Headaches
 - Weakness or numbness: facial or extremity, unilateral or bilateral
 - Seizure activity
 - Syncope
 - Hearing or visual changes
 - Speech difficulties
- **Cardiovascular status assessment and monitoring(23)**
 - Complete assessments, including:
 - Hemodynamic stability
 - Blood pressure
 - Heart rate and rhythm, including arrhythmias or palpitations
 - Telemetry
 - Presence of implantable cardiac defibrillator or pacemaker
 - Monitor for:
 - Chest pain or angina
 - Cyanosis
 - Fatigue
 - Perfusion deficiency, particularly to lower extremities (eg, pedal pulses, color, temperature, or sensation of extremities)
 - Weight gain and dependent edema
 - Infusion therapy needs
 - Transfusions with packed red blood cells or platelets
 - Presence of vascular access device
- **Respiratory status assessment and monitoring(8)(21)**
 - Complete assessments, including:
 - Lung sounds
 - Crackles
 - Rhonchi
 - Wheezes
 - Ability to cough and deep breathe without restriction due to pain
 - Respiratory rate, rhythm, quality, and effort, including:
 - Cough and characteristics of sputum
 - Oxygen saturation by pulse oximetry
 - Peak expiratory flow rate

- Intubation and ventilation parameters
- Use of supplemental oxygen
- Monitor for:
 - Respiratory status changes (eg, drop in pulse oximetry, sputum, peak flow rate)
 - Cyanosis
 - Orthopnea
 - Dyspnea
 - Intubation and ventilation parameter changes
 - Need for suctioning, chest physiotherapy, or nebulizers to improve pulmonary status
- **Gastrointestinal status assessment and monitoring(8)(21)**
 - Complete assessments, including:
 - Fluid or electrolyte imbalance
 - Weight and body mass index
 - Bowel function, sounds, and movements: quantity, color, frequency, and consistency
 - Nausea and vomiting
 - Abdominal pain, tenderness, distention, or rigidity
 - Nutritional status(14):
 - Risk for malnutrition
 - Ability to tolerate oral fluids and foods
 - Fluid or electrolyte imbalance
 - Enteral or parenteral nutrition
 - Weight and body mass index
 - Risk for dehydration
 - Preoperative and postoperative status, including NPO (time frame prescribed by ordering physician) before surgery and possible need for nutritional consultation(15)
 - Monitor for:
 - Possible complications, which may include hemorrhage, bile duct injury, or anastomotic leaks
 - Abdominal pain, distention, ascites, tenderness, or masses
 - Nausea or vomiting
 - Jaundice
 - Bowel function changes (eg, constipation, diarrhea, melena)
- **Renal status assessment and monitoring(8)**
 - Complete assessments, including:
 - Renal function (eg, intake and output, creatinine clearance)
 - Urinary appearance and output
 - Pain or burning with urination
 - Urgency or hesitancy
 - Hematuria or urine discoloration
 - Stones retrieved on straining urine
 - Urethral catheter, if present
 - Monitor for:
 - Renal function changes (eg, intake and output, creatinine clearance)
 - Flank pain (with or without radiation to groin)
 - Urinary tract infection
 - Chills
 - Fever
 - Dysuria
 - Frequency
 - Urinary tract obstruction
 - Frequent urination
 - Oliguria or anuria
 - Weak stream
 - Nephrostomy drainage for ureteral obstruction
 - Catheter care and management (ie, Foley, suprapubic, or straight)
- **Integumentary status assessment and monitoring(8)**
 - Complete assessments, including:
 - Skin status, including cyanosis, jaundice, diaphoresis, or poor skin turgor
 - Skin integrity, including inflammation, rash, open wounds, pressure injuries, or bruising
 - Lesions
 - Wound and drain assessment
 - Incision for healing, swelling, drainage, odor, and other signs of infection
 - Monitor wound site, wound bed, size and depth. See Wounds Due to Surgery, Injury, or Pressure  SR.

- Drain monitoring for change in color, consistency, and amount of drainage, or integrity of skin surrounding drain
- Dressings, including signs of skin breakdown
- Monitor for:
 - Skin status or integrity changes
 - Signs of skin or wound infection
 - Redness, warmth, swelling, tenderness, or drainage
 - Cellulitis - acute, nonnecrotizing inflammation of dermis and hypodermis(18)
 - Abscesses and exudates
 - Necrotizing fasciitis involves deep soft tissues, causing increasingly severe localized pain.
- **Obstetric status assessment and monitoring(24)(25)**
 - Complete assessments, including:
 - Fundus firm at umbilicus
 - Vaginal discharge amount, color, and consistency
 - Birth route wound and suture site if cesarean birth or episiotomy
 - Breast or formula feeding
 - Monitor for:
 - Wound complications
 - Pain changes or complaints of new pain
 - Vaginal discharge changes
 - Breast-feeding complications (eg, mother's breast engorgement, difficulty latching onto breast)
 - Postpartum depression
- **Cast care(26)(27)**
 - Complete assessments, including:
 - Condition of cast and any pressure points on skin
 - Skin integrity around cast
 - Impact on ADL or IADL performance
 - Pin sites if external fixation device
 - Monitor for:
 - Compartment syndrome signs and symptoms
 - Hot spots near cast that may indicate infection of wound
 - Drainage from inside cast: color, consistency, and odor
 - Neurovascular compromise
 - Cast breakdown or loosening
 - Increased pain not relieved by elevation or medication
- **Discharge planning.** See Discharge Planning Tool  SR.(20)
 - Discharge planning initiated
 - Discharge planning completed

Pediatric

- **General status assessment and monitoring** changes based on age(28)
 - Infants
 - Height, weight, and head circumference appropriate for age
 - Developmental stage, using standardized screening tool for infant development
 - Overall clinical appearance (eg, mildly ill, toxic)
 - Pain, which is dependent on caregivers to assess pain and to determine effectiveness of management efforts. Examples of measurement tools include(29):
 - CRIES is an infant postoperative pain scale assessing 5 variables (C-crying, R-requires oxygen, I-increased vital signs, E-expression, S-sleeplessness) rated on a 0 to 2 point scale for each variable (much like an APGAR score).
 - FLACC is a clinical observational tool that includes 5 categories of behavior (F-facial expression, L-leg movement, A-activity, C-cry, and C-consolability) rated on a 0 to 2 point scale for each category.
 - Fever
 - Irritability and excessive, prolonged crying
 - Child or adolescent
 - Height and weight appropriate for age[C]
 - Activity and developmental status appropriate for age
 - Fever
 - Pain assessment and management[D]
 - Child measurement tools include:
 - Visual Analog Scale (VAS) for children 8 years of age and older
 - Faces scale or Color Analog Scale for children 4 years of age and older
 - Autonomic measures (eg, heart rate, blood pressure, heart rate spectral analyses)

- Adolescent measurement tools include:
 - Adolescent Pediatric Pain Tool (APPT)
 - Pediatric Pain Questionnaire (PPQ)
 - Autonomic measures (eg, heart rate, blood pressure, heart rate spectral analyses)
 - Responses to hospitalization and surgical process include(30):
 - Infant stress (eg, from separation, pain) responses include:
 - Protest (crying, screaming)
 - Despair
 - Detachment
 - Toddler stress (eg, from separation, pain, fear, loss of control) responses include:
 - Protest (crying, screaming, clinging to parent)
 - Physical resistance and retaliation (eg, kicking, biting, hitting)
 - Despair
 - Detachment or withdrawal
 - Regression
 - Preschool children stress (eg, from separation, pain, fear, loss of control) responses include:
 - Protest (refusing to eat or drink, clinging to parent)
 - Physical resistance and retaliation (eg, attempting to escape, breaking toys)
 - Despair
 - Detachment or withdrawal
 - Regression
 - School-age children stress (eg, from separation, pain, fear, loss of control) responses include:
 - Regression
 - Depression, hostility, or frustration
 - Resistance (eg, attempting to escape, plea bargaining)
 - Adolescent stress (eg, from separation -- particularly from peer group, pain, fear, loss of control) responses include:
 - Irritability and aggression toward parents or caregivers
 - Rejection or jealousy of siblings
 - Rejection, uncooperativeness, or withdrawal
 - Self-assertion, anger, or frustration
 - Isolation from peers
 - ADL and IADL ability assessed. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#).
 - Psychosocial status assessed. See Psychosocial Assessment [SR](#).
 - Social determinants of health assessed. See Social Determinants of Health Screening Tool [SR](#).
 - Medication reconciliation completed. See Medication Reconciliation Tool [SR](#).
- **Neurologic status assessment and monitoring**
 - Infants
 - Alertness appropriate for age
 - Level of consciousness
 - Fontanel and head circumference
 - Oral motor function
 - Ability to hear
 - Muscle tone (ie, hypertonia or hypotonia) and reflexes
 - Child or adolescent
 - Cognitive function
 - Concentration
 - Headache
 - Level of consciousness and alertness appropriate for age
 - Motor function and possible deficits
 - Tremors or seizure activity
 - Speech and swallowing ability
 - Visual disturbances or changes
- **Cardiac status assessment and monitoring(31)**
 - Infants
 - Color, temperature (eg, pallor, cyanosis, mottling)
 - Capillary refill
 - Blood pressure by Doppler
 - Heart rate and rhythm
 - Heart murmurs or defects
 - Child or adolescent

- Blood pressure
- Heart rate and rhythm, including new or changed murmur
- Change in temperature, color, or sensation of distal extremities
- Peripheral pulses
- Peripheral edema
- **Respiratory status assessment and monitoring(32)(33)**
 - Infants
 - Lung sounds
 - Respiratory rate, rhythm, and effort, including signs of respiratory distress (eg, grunting, nasal flaring, retractions, wheezing)
 - Oxygen saturation
 - Episodes of apnea and readings from apnea monitor
 - Child or adolescent
 - Lung sounds
 - Respiratory rate, rhythm, and effort
 - Dyspnea
 - Wheezing
 - Cough, productive or dry, including characteristics of sputum
- **Gastrointestinal status assessment and monitoring(28)**
 - Infants
 - Growth and weight for age
 - Feeding patterns and ability to suck or swallow
 - Breast-fed
 - Bottle-fed
 - Fluid and nutrition intake, including feeding problems (eg, spitting up, choking, gagging)
 - Quality of bowel sounds and bowel movements (eg, quantity, color, frequency, consistency)
 - Urine output, including amount, frequency, and color
 - Child or adolescent
 - Growth and weight for age
 - Feeding pattern
 - Ability to tolerate food and fluids
 - Appetite
 - Abdominal pain or distention
 - Nausea or vomiting
 - Quality of bowel sounds and bowel movements: quantity, color, frequency, and consistency
 - Urine output, including amount, frequency, and color
 - For all age groups:
 - Assess and support normal play and other activities as allowed.
 - Assess signs and symptoms indicating lack of nutrition or signs of complications.
- **Integumentary status assessment and monitoring(28)**
 - Infants
 - Skin color, temperature, and turgor
 - Rashes or irritation in diaper area
 - Open wounds, drainage, or odor
 - Yellow sclera, conjunctiva, oral mucosa, and skin
 - Child or adolescent
 - Rash
 - Abrasions or contusions
 - Open wounds, drainage, or odor
 - Soft tissue damage
 - Abscess formation
 - Burns
- **Behavior problems**, child or adolescent
 - Nonadherence
 - Inattention
 - Aggression
- **Discharge planning**. See Discharge Planning Tool  SR.(20)
 - Discharge planning initiated
 - Discharge planning completed

Behavioral Health

- **Assessments(34)(35)**

- Psychological, cognitive, and behavioral health symptoms
 - Affect and mood
 - Orientation and memory
 - Thought, content, and process (eg, confusion, hallucinations, or delusions)
 - Coping measures utilized
 - Anxiety
 - Hopelessness
 - Low self-esteem
 - Poor judgment
 - Pressured speech
 - Suicidal potential
 - Psychosocial status. See Psychosocial Assessment  SR.(22)
- Physical behavior assessment(22)
 - Psychomotor agitation or retardation
 - Aggression
 - Ability to communicate
 - Self-management abilities
 - Medication and dietary adherence
 - Bizarre behavior patterns (eg, motor agitation and restlessness, or stupor)
 - Risk-taking behaviors (eg, criminal behaviors or hazardous activities)
 - Self-mutilation
 - Use of substances of abuse
- Social functioning, including ability to work or attend school(22)
 - Impact of symptoms on family dynamics
 - Diversional or recreational behavior
 - Formal and informal support systems
 - Social isolation
- ADL and IADL ability assessed. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment  SR.
- Medication reconciliation completed. See Medication Reconciliation Tool  SR.

- **Monitoring**

- Behavioral health pattern changes(22)
 - Eating patterns
 - Sleeping patterns
 - Personal hygiene
 - Elimination patterns
 - Level of interest in sex
- Physical behavior status changes
- Psychosocial status changes

- **Discharge planning.** See Discharge Planning Tool SR.(36)

- Discharge planning initiated
- Discharge planning completed

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Footnotes

[A] Malnutrition may be associated with higher rates of complications, increased nosocomial infections, higher hospital costs, higher mortality, and longer length of hospital stay (LOS). Complications of malnutrition increase LOS, therapeutic hospital cost, and ultimately the cost of patient rehabilitation.(15) [A in Context Link 1]

[B] Systemic diseases (eg, diabetes, rheumatoid arthritis, syphilis, muscular dystrophy) have ocular manifestations. A history of cardiac or pulmonary disease is important because beta-adrenergic blockers are often used to treat glaucoma.(19) [B in Context Link 1]

[C] Head circumference may be measured in children up to 36 months of age or older if size is questionable.(28) [C in Context Link 1]

[D] Measurement of pain in children may be influenced by limited cognitive or language skills or by positive or negative consequences of pain reports or behaviors.(29) [D in Context Link 1]

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